WORKED SOLUTION

STEP III 1997, Question 14

Probability and Statistics • Products of Random Variables

STEP III 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q14

The key idea

The perimeter is a linear function of the length and breadth, so its mean and variance come straight from the rules for sums of independent variables. The area is a product, and for independent variables the expectation of a product is the product of the expectations; applying this to LB and to L^2B^2 gives the mean and variance. To show dependence, the cleanest route is to find a non-zero covariance, because independent variables always have zero covariance.

Solution

Let L and B be the length and breadth of a randomly chosen plate, so that L and B are independent with

$$\mathrm{E}(L) = \mu_1, \quad \mathrm{Var}(L) = \sigma_1^2, \qquad \mathrm{E}(B) = \mu_2, \quad \mathrm{Var}(B) = \sigma_2^2.$$

Let $P = 2(L + B)$ be the perimeter and $A = LB$ the area. We shall use $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$ repeatedly, so that $\mathrm{E}(L^2) = \mu_1^2 + \sigma_1^2$ and $\mathrm{E}(B^2) = \mu_2^2 + \sigma_2^2$.

The perimeter

By linearity of expectation,

$$\mathrm{E}(P) = 2(\mu_1 + \mu_2).$$

Since L and B are independent, the variance of their sum is the sum of their variances, so

$$\mathrm{Var}(P) = 2^2(\mathrm{Var}(L) + \mathrm{Var}(B)) = 4(\sigma_1^2 + \sigma_2^2), \qquad \text{s.d.}(P) = 2\sqrt{\sigma_1^2 + \sigma_2^2}.$$

The area

Because L and B are independent, $\mathrm{E}(LB) = \mathrm{E}(L)\,\mathrm{E}(B)$, so

$$\mathrm{E}(A) = \mu_1\mu_2.$$

For the variance we need $\mathrm{E}(A^2) = \mathrm{E}(L^2B^2)$.

Ask yourself Can we still split $\mathrm{E}(L^2B^2)$ into a product?

Yes: functions of independent variables are independent, so L^2 and B^2 are independent and $\mathrm{E}(L^2B^2) = \mathrm{E}(L^2)\,\mathrm{E}(B^2)$. Therefore

$$\begin{aligned}\mathrm{Var}(A) &= \mathrm{E}(L^2)\,\mathrm{E}(B^2) - \mu_1^2\mu_2^2 \\ &= (\mu_1^2+\sigma_1^2)(\mu_2^2+\sigma_2^2) - \mu_1^2\mu_2^2 \\ &= \sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2,\end{aligned}$$

and

$$\mathrm{s.d.}(A) = \sqrt{\sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2}.$$

Notice that the variance of the product is *not* just $\sigma_1^2\sigma_2^2$: the spread in each dimension is magnified by the typical size of the other.

Perimeter and area are not independent

If P and A were independent we would have $\mathrm{E}(PA) = \mathrm{E}(P)\,\mathrm{E}(A)$, that is, $\mathrm{Cov}(P, A) = 0$. We compute the covariance. First,

$$\mathrm{E}(PA) = 2\,\mathrm{E}(L^2B) + 2\,\mathrm{E}(LB^2).$$

Using independence again (now of L^2 and B, and of L and B^2),

$$\mathrm{E}(L^2B) = (\mu_1^2+\sigma_1^2)\mu_2, \qquad \mathrm{E}(LB^2) = \mu_1(\mu_2^2+\sigma_2^2).$$

Hence

$$\begin{aligned}\mathrm{Cov}(P, A) &= \mathrm{E}(PA) - \mathrm{E}(P)\,\mathrm{E}(A) \\ &= 2[(\mu_1^2+\sigma_1^2)\mu_2 + \mu_1(\mu_2^2+\sigma_2^2)] - 2(\mu_1+\mu_2)\mu_1\mu_2 \\ &= 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2).\end{aligned}$$

Lengths and breadths are positive, so $\mu_1 > 0$ and $\mu_2 > 0$; the standard deviations are given to be non-zero, so $\sigma_1^2 > 0$ and $\sigma_2^2 > 0$. Thus $\mathrm{Cov}(P, A) > 0$.

Since independent variables have zero covariance, P and A cannot be independent. This makes sense: a plate that is larger than average in either direction tends to have both a larger perimeter and a larger area.

Ask yourself If the covariance had come out as zero, would that have shown P and A were independent?

No. The implication only runs one way: independence forces zero covariance, but zero covariance does not force independence. So a *non-zero* covariance proves dependence, which is exactly what we have; a zero covariance would have told us nothing.

Checks

- Take L and B independent, each equal to 1 or 3 with probability $\frac{1}{2}$. Then $\mu_i = 2$, $\sigma_i^2 = 1$ and the formula gives $\mathrm{Cov}(P, A) = 2(2 + 2) = 8$, which matches direct enumeration of the four equally likely plates. Dependence is also visible directly: $\mathrm{P}(P = 4) = \mathrm{P}(A = 1) = \frac{1}{4}$, but $\mathrm{P}(P = 4 \text{ and } A = 1) = \frac{1}{4} \neq \frac{1}{16}$, since both events say $L = B = 1$.
- A simulation with $\mu_1 = 5$, $\mu_2 = 3$, $\sigma_1 = 0.7$, $\sigma_2 = 0.4$ gave $\text{s.d.}(A) \approx 2.919$ against the formula's 2.913, and $\mathrm{Cov}(P, A) \approx 4.56$ against 4.54.
- If $\sigma_2 \to 0$ (fixed breadth μ_2), $\text{s.d.}(A) \to \mu_2\sigma_1$, as it should for $A = \mu_2 L$.

Common slips

- Writing $\mathrm{Var}(LB) = \sigma_1^2\sigma_2^2$, or $\mathrm{Var}(LB) = \mathrm{Var}(L)\mathrm{Var}(B) + \ldots$ with terms missing; always go through $\mathrm{E}(L^2B^2)$.
- Forgetting to square the factor 2 in $\mathrm{Var}(2(L + B))$.
- Arguing that P and A are dependent "because both depend on L and B". That is intuition, not proof; the covariance calculation (or an explicit example of unequal probabilities) is what settles it.

Beyond STEP

The distinction between uncorrelated and independent random variables is central in first-year probability: independence means the joint distribution factorises, which is much stronger than a single moment condition. The calculation for $\mathrm{Var}(LB)$ is also the starting point for the "propagation of errors" formulas used throughout experimental science. Grimmett and Welsh, *Probability: An Introduction*, treats independence, covariance and products of random variables carefully.

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